

Arduino vs Raspberry Pi

Ever since the Raspberry Pi was announced, it has been a topic of much debate online. Starting from the radically different naming schemes—masculine Italian name vs sweet fruit pie—Arduino and Raspberry Pi boards are significantly different in a lot of things. For one, the Arduino is an Open Source single-board microcontroller, while the Raspberry Pi is a single-board computer.

Ask any electronics enthusiast and he will tell you why it is always better to get the Arduino if this is your first time dabbling with electronics. On the other hand, if you have already dipped your toes in this field, you could go ahead with the Raspberry Pi.

The table compares the two boards. It should help you make a good decision if you are new to this field.

Arduino vs Raspberry Pi

Arduino	Raspberry Pi
Extremely easy to get working	Less easier to get working
A typical embedded system with easy-to-develop software	Complete computer running Linux
Support is available almost everywhere	Support is currently limited but should increase over time
More suitable for controlling hardware (robotics)	Features an extremely powerful GPU and can handle hi-def content
Umpteen, different kits are available	Very few kits are available at present
Low power consumption, capable of even microamps with very low clock	Comparatively higher power consumption than the Arduino
Verdict: While the Raspberry Pi is a great high-performance option, Arduino is a good general-purpose board.	

and uses the ARM Cortex A8 processor clocked at 600 MHz. The board is just 7.6x7.6 cm² (3x3 inch²) in size and comes coloured in what I like to call as the TI Red. It can run Linux, Risc OS, Symbian, Nokia's Maemo OS and Android. The board also supports OpenGL ES 2.0 through its PowerVR SGX530 GPU.

The BeagleBoard comes in BeagleBoard-xM and BeagleBone versions. Up to four expansion boards, called capes, can be added to the BeagleBoard. Other expansion boards include the BeagleJuice—a Li-ion battery pack.

The BeagleBone is a stripped-down version of the BeagleBoard. It is amazingly compact at 8.6x5.3 cm² (3.4x2.1 inch²) and sold online for ₹ 6560. On the other hand, the BeagleBoard XM is available for ₹ 10,920.

STMicroelectronics' STM32 Discovery

This board from the STMicroelectronics stable is yet another inexpensive development board that runs on ARM Cortex-M3 prototyping platform. The STM32 Discovery line comes in STM32L and STM32VL versions.

The STM32L Discovery is an ultra-low-power discovery board based on the STM32L core [STM32L1xxx] and includes an ST-Link/V2 embedded debugging tool interface, an LCD (24 segments,

four commons), LEDs, pushbuttons and a linear touch sensor. Non-intrusive debug is possible with the in-circuit debugger available on STM32 Value Line Discovery. The kit can also be used as an ST-link for your own board.

The board's power supply can be either through the USB bus or an external 3.3V or 5V supply. The STM32 is also said to be one of the lowest power-consuming, 32-bit, 72MHz computer in the market today. It costs around ₹ 940 in India.

If you are new to STM32 Discovery boards, you can check out numerous examples based on this board at www.st.com/stm32-discovery.

The STM32VL Value Line board is priced at ₹ 865 in India. Unlike the Discovery kit, it is based on the STM32F core (STM32F1xxx). The STM32L is meant for ultra-low-power devices and hence uses a different core.

Keep in mind that the STM32L and STM32VL are different families of boards. These are not as compatible as their almost similar names suggest. Debugging is different for both these boards.

TI MSP430 Launchpad

Unveiled in 2010, this board is a development platform for Texas Instruments' MSP430 line of low-cost microcontrollers. It is a mixed-signal integrated circuit based development board that costs just ₹ 393. The kit features built-in flash emulation for debugging and pro-

gramming and a 14- or 20-pin dual-inline package (DIP) socket. It also comes with ten general-purpose input-output (GPIO) pins. GPIO pins are generic pins. Their behaviour can be controlled through software.

"The USP for the MSP430 is ultra-low power consumption (TI's online document shows that at 0.1 per cent active, an MSP430 can last for over 20 years on a 200mAh coin cell). It is used as an ultra-low-power MCU in niche devices such as blood group meters, general metering, home automation and wireless smoke detectors. We do have success with other architectures as well, but 70 per cent of our success can be attributed to the MSP430. That's why we have also launched the low-cost Launchpad based on the MSP430, which sells for \$4.30. It has all the debug capabilities and one can do all the prototyping on the Launchpad itself," says Shailesh Thakurdesai, general manager-business development, Texas Instruments.

Each Launchpad board comes bundled with some extras. In addition to the onboard 32.768kHz clock crystal, you get half-a-metre USB cable, two-pin headers and two-pin sockets for the pin breakout pads, two different MSP430s (MSP430G2553 and MSP430G2452), and two free integrated development environments (Code Composer Studio 4 and IAR Embedded Workbench Kickstart). Note that